

Calculadora BMC / Wolfboard — Referencia técnica del sistema

Documento auto-generado por `scripts/generate-system-reference.mjs` a partir del código real del repo. **No editar a mano** — se regenera (ver workflow `product-docs`). Es la representación *fiel y cableada* de cómo está construido el sistema; el recorrido visual de la UI está en `PRODUCT-OVERVIEW.md`.

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Versión	calculadora-bmc v3.1.5
Commit	7053c13 (<code>claude/product-tour-doc-setup-jptsox</code>)

Arquitectura (resumen)

- **Frontend:** React 18 + Vite (SPA, `:5173` en dev). Entry `src/App.jsx` (router con 27 rutas).
- **API:** Express 5 sobre Node (ES modules, `:3001`). Entry `server/index.js`; expone 333 rutas en 41 módulos de `server/routes/`.
- **Datos:** PostgreSQL (`pg`) + pgvector; Google Sheets (service account); integraciones MercadoLibre, WhatsApp Cloud, GCS, OpenAI/Anthropic. Ver §Integraciones.
- **Deploy:** Frontend en Vercel; API en Google Cloud Run (`panelin-calc`).

1. Superficie de API (333 rutas en 41 módulos)

`activity.js` — montado en `(raíz)`

Método	Ruta
GET	<code>/api/activity/status</code>
GET	<code>/api/activity/today</code>
GET	<code>/api/activity/buckets</code>

`agentChat.js` — montado en `/api`

Método	Ruta
GET	<code>/api/agent/ai-options</code>
GET	<code>/api/agent/tool-stats</code>
GET	<code>/api/agent/tools-manifest</code>
POST	<code>/api/agent/exec-tool</code>
POST	<code>/api/agent/chat</code>

agentConversations.js — montado en /api

Método	Ruta
GET	/api/agent/stats
GET	/api/agent/conversations
GET	/api/agent/conversations/weekly-digest
POST	/api/agent/conversations/analyze-batch
GET	/api/agent/conversations/:id
GET	/api/agent/conversations/:id/analysis

agentFeedback.js — montado en /api

Método	Ruta
POST	/api/agent/feedback
GET	/api/agent/feedback
GET	/api/agent/feedback/stats

agentTraining.js — montado en /api

Método	Ruta
POST	/api/agent/train
PUT	/api/agent/train/:id
DELETE	/api/agent/train/bulk
PATCH	/api/agent/train/bulk
DELETE	/api/agent/train/:id
GET	/api/agent/training-kb
GET	/api/agent/training-kb/match
GET	/api/agent/dev-config
POST	/api/agent/dev-config
POST	/api/agent/prompt-preview
POST	/api/agent/training/log-event
GET	/api/agent/dev-config/:section/history
POST	/api/agent/dev-config/:section/revert
POST	/api/agent/knowledge/clear-cache
GET	/api/agent/training-kb/score-config
POST	/api/agent/training-kb/score-config
POST	/api/agent/autolearn
GET	/api/agent/autolearn/pending
POST	/api/agent/autolearn/:id/approve

Método	Ruta
POST	/api/agent/autolearn/:id/reject
GET	/api/agent/training-kb/health
POST	/api/agent/training-kb/:id/mark-reviewed
POST	/api/agent/training-kb/import
GET	/api/agent/training-kb/conflicts
POST	/api/agent/training-kb/:id/resolve-conflict
POST	/api/agent/training-kb/generate-ml-overrides
GET	/api/agent/training-kb/analytics

agentTranscribe.js — montado en /api

Método	Ruta
POST	/api/agent/transcribe

agentVoice.js — montado en /api

Método	Ruta
POST	/api/agent/voice/session
POST	/api/agent/voice/action
GET	/api/agent/voice/errors
POST	/api/agent/voice/errors/clear
GET	/api/agent/voice/health

aiAnalytics.js — montado en /api

Método	Ruta
GET	/api/ai-analytics/trends

authGoogle.js — montado en /api

Método	Ruta
POST	/api/auth/google
POST	/api/auth/refresh
POST	/api/auth/logout
GET	/api/auth/me
GET	/api/auth/me/grants

authMfa.js — montado en (raíz)

Método	Ruta
POST	/auth/mfa/enroll
POST	/auth/mfa/verify
POST	/auth/mfa/disable
POST	/auth/mfa/challenge

bmcDashboard.js — montado en /api

Método	Ruta
GET	/api/cotizaciones
GET	/api/proximas-entregas
GET	/api/coordinacion-logistica
GET	/api/audit
GET	/api/pagos-pendientes
GET	/api/metas-ventas
GET	/api/calendario-vencimientos
GET	/api/ventas
GET	/api/ventas/tabs
GET	/api/stock-ecommerce
GET	/api/stock-kpi
GET	/api/productos-maestro
GET	/api/productos-maestro/reconcile
GET	/api/productos-maestro/links
PUT	/api/productos-maestro/links
POST	/api/productos-maestro/push
GET	/api/kpi-financiero
GET	/api/stock/history
GET	/api/kpi-report
GET	/api/fiscal/bps-irae
POST	/api/cotizaciones
PATCH	/api/cotizaciones/:id
POST	/api/pagos
PATCH	/api/pagos/:id
POST	/api/ventas
PATCH	/api/stock/:codigo
POST	/api/marcar-entregado
GET	/api/actualizar-precios-calculadora

Método	Ruta
POST	/api/crm/suggest-response
POST	/api/crm/parse-email
POST	/api/crm/ingest-email
POST	/api/crm/parse-conversation
POST	/api/ventas/logistica-fecha-entrega
POST	/api/matriz/push-pricing-overrides
GET	/api/email/panelsim-summary
POST	/api/email/draft-outbound
GET	/api/crm/cockpit/row/:rowNum
POST	/api/crm/cockpit/quote-link
POST	/api/crm/cockpit/approval
POST	/api/crm/cockpit/mark-sent
POST	/api/crm/cockpit/taxonomy-row
POST	/api/crm/cockpit/save-response
POST	/api/crm/cockpit/send-approved
GET	/api/crm/cockpit/ml-queue
GET	/api/crm/cockpit/wa-queue
POST	/api/crm/cockpit/sync-ml
GET	/api/crm/cockpit/unified-queue
GET	/api/consultations
POST	/api/consultations/:consultationId/reply
POST	/api/crm/cockpit/sync-all

bugs.js — montado en /api/bugs

Método	Ruta
POST	/api/bugs/report
GET	/api/bugs/

calc.js — montado en /calc

Método	Ruta
GET	/calc/openapi
GET	/calc/gpt-entry-point
POST	/calc/interaction-log
GET	/calc/interaction-log/list
GET	/calc/interaction-log/file/:name

Método	Ruta
POST	/calc/cotizar/presupuesto-libre
POST	/calc/cotizar
POST	/calc/cotizar/pdf
GET	/calc/pdf/:id
GET	/calc/cotizaciones
GET	/calc/cotizaciones/:id
POST	/calc/cotizaciones/:id/cancelar
GET	/calc/catalogo
GET	/calc/escenarios
GET	/calc/informe

clientes/customers.js — montado en (raíz)

Método	Ruta
GET	/api/clientes/customers
GET	/api/clientes/customers/summary

clientes/followups.js — montado en (raíz)

Método	Ruta
POST	/api/clientes/followups
GET	/api/clientes/followups

deepResearch.js — montado en /api

Método	Ruta
POST	/api/research/deep
GET	/api/research/deep/:id
POST	/api/research/deep/:id/cancel

followups.js — montado en /api

Método	Ruta
GET	/api/followups
POST	/api/followups
GET	/api/followups/:id
PATCH	/api/followups/:id
POST	/api/followups/:id/done
POST	/api/followups/:id/snooze

Método	Ruta
DELETE	/api/followups/:id

identityAdmin.js — montado en (raíz)

Método	Ruta
GET	/api/admin/users
GET	/api/admin/users/:id
POST	/api/admin/users/:id/role-grants
DELETE	/api/admin/users/:id/role-grants/:role
PATCH	/api/admin/users/:id/module-grants/:module
POST	/api/admin/users/:id/suspend
POST	/api/admin/users/:id/reactivate
POST	/api/admin/users/:id/revoke-sessions

identityAnalytics.js — montado en (raíz)

Método	Ruta
GET	/api/admin/analytics/active-users
GET	/api/admin/analytics/module-usage
GET	/api/admin/analytics/error-rate
GET	/api/admin/analytics/timeline
GET	/api/admin/analytics/top-actions

identityMe.js — montado en (raíz)

Método	Ruta
GET	/api/me/notifications
PATCH	/api/me/notifications/:id
POST	/api/access-requests
GET	/api/me/access-requests
GET	/api/admin/access-requests
PATCH	/api/admin/access-requests/:id
POST	/api/me/special-quote-requests
GET	/api/me/special-quote-requests
GET	/api/me/quotes
GET	/api/me/quotes/:id
POST	/api/me/quotes
DELETE	/api/me/quotes/:id

Método	Ruta
POST	/api/me/quotes/claim
GET	/api/admin/sheets/clientes/status
POST	/api/admin/sheets/clientes/reconcile
POST	/api/admin/sheets/clientes/sync/:quote_id
POST	/api/me/activity
GET	/api/me/activity
GET	/api/me/users/search
GET	/api/me/threads
GET	/api/me/threads/:id/messages
POST	/api/me/threads
POST	/api/me/threads/:id/messages
PATCH	/api/me/threads/:id/read

internal/presupOrchestrator.js — montado en /api/internal/presup

Método	Ruta
POST	/api/internal/presup/run
GET	/api/internal/presup/status
GET	/api/internal/presup/run/example

LegacyQuote.js — montado en /

Método	Ruta
GET	/ready
POST	/find_products
POST	/resolve_product
POST	/product_price
POST	/check_availability
POST	/calculate_quote
POST	/calculate_quote_v2

marketing.js — montado en /api/marketing

Método	Ruta
GET	/api/marketing/dashboard/summary
GET	/api/marketing/dashboard/competitors
GET	/api/marketing/dashboard/alerts
GET	/api/marketing/mystery-shopping

Método	Ruta
PATCH	/api/marketing/mystery-shopping/:id/status
POST	/api/marketing/etl/run

mlEtlRun.js — montado en (raíz)

Método	Ruta
POST	/api/ml/etl-run
GET	/api/ml/etl-run/latest
GET	/api/ml/etl-run/:id

mlSearch.js — montado en (raíz)

Método	Ruta
GET	/api/ml/search

panelinInternal.js — montado en /api/internal/panelin

Método	Ruta
GET	/api/internal/panelin/whoami
GET	/api/internal/panelin/tools
GET	/api/internal/panelin/policies
POST	/api/internal/panelin/invoke

pdf.js — montado en /api/pdf

Método	Ruta
POST	/api/pdf/generate
GET	/api/pdf/metrics

planInterpret.js — montado en /api

Método	Ruta
POST	/api/plan/interpret

quoteExport.js — montado en (raíz)

Método	Ruta
POST	/api/admin/export
GET	/api/me/quotes/:id/export.json
GET	/api/me/quotes/:id/export.csv
GET	/api/me/quotes/:id/export.pdf

Método	Ruta
GET	/api/me/quotes/:id/export.html

quotes.js — montado en /api

Método	Ruta
GET	/api/quotes/counter
POST	/api/quotes/counter/next

shopify.js — montado en (raíz)

Método	Ruta
GET	/auth/shopify
GET	/auth/shopify/callback
POST	/webhooks/shopify
GET	/api/shopify/products
GET	/api/shopify/catalog/full
GET	/admin/questions
POST	/admin/answer
POST	/admin/auto-config
GET	/admin/auto-config

superAgent.js — montado en /api/agent

Método	Ruta
POST	/api/agent/quote-lead

tasks.js — montado en /api/tasks

Método	Ruta
GET	/api/tasks/lists
GET	/api/tasks/lists/:id
GET	/api/tasks/lists/:id/tasks
GET	/api/tasks/lists/:id/tasks/:taskId
POST	/api/tasks/lists
DELETE	/api/tasks/lists/:id
POST	/api/tasks/lists/:id/tasks
PATCH	/api/tasks/lists/:id/tasks/:taskId
DELETE	/api/tasks/lists/:id/tasks/:taskId

tasksOAuth.js — montado en /auth/tasks

Método	Ruta
GET	/auth/tasks/init
GET	/auth/tasks/callback
GET	/auth/tasks/scope-probe
POST	/auth/tasks/revoke

tasksSync.js — montado en /sync

Método	Ruta
POST	/sync/google-tasks/pull

teamAssist.js — montado en /api/team-assist

Método	Ruta
GET	/api/team-assist/health
GET	/api/team-assist/agents
POST	/api/team-assist/chat

traktime.js — montado en (raíz)

Método	Ruta
GET	/api/traktime/health
GET	/api/traktime/me
GET	/api/traktime/clients
POST	/api/traktime/clients
PATCH	/api/traktime/clients/:id
GET	/api/traktime/projects
POST	/api/traktime/projects
PATCH	/api/traktime/projects/:id
GET	/api/traktime/projects/:id/members
POST	/api/traktime/projects/:id/members
DELETE	/api/traktime/projects/:id/members/:userId
GET	/api/traktime/tasks
POST	/api/traktime/tasks
GET	/api/traktime/tags
POST	/api/traktime/tags
GET	/api/traktime/reports/billable
GET	/api/traktime/invoices
POST	/api/traktime/invoices/draft

Método	Ruta
POST	/api/traktime/invoices/:id/issue
POST	/api/traktime/invoices/:id/mark-paid
POST	/api/traktime/invoices/:id/void
GET	/api/traktime/invoices/:id/pdf
POST	/api/traktime/admin/mirror-now
GET	/api/traktime/timer/current
POST	/api/traktime/timer/start
POST	/api/traktime/timer/stop
GET	/api/traktime/entries
POST	/api/traktime/entries
PATCH	/api/traktime/entries/:id
DELETE	/api/traktime/entries/:id
GET	/api/traktime/reports/summary
GET	/api/traktime/day-report
GET	/api/traktime/month-report

transportista.js — montado en /api

Método	Ruta
GET	/api/transportista/health
POST	/api/trips
POST	/api/trips/:trip_id/confirm
POST	/api/trips/:trip_id/assign
POST	/api/trips/:trip_id/driver-link/regenerate
GET	/api/trips/:trip_id/timeline
GET	/api/trips/:trip_id/state
POST	/api/trips/:trip_id/close
GET	/api/driver/trips
GET	/api/driver/trips/:trip_id
POST	/api/driver/events
POST	/api/driver/evidence/upload-url
POST	/api/driver/evidence/commit
POST	/api/driver/evidence/upload-b64

wa.js — montado en /api

Método	Ruta
GET	/api/wa/config
PATCH	/api/wa/settings
PATCH	/api/wa/flags/:key
POST	/api/wa/settings/test-ai
GET	/api/wa/operators
POST	/api/wa/operators/invite
DELETE	/api/wa/operators/:id/sessions
GET	/api/wa/rules
POST	/api/wa/rules/preview
GET	/api/wa/webhooks
POST	/api/wa/webhooks/:id/test
GET	/api/wa/audit-log
POST	/api/wa/auth/request-magic-link
GET	/api/wa/auth/verify
POST	/api/wa/auth/verify
POST	/api/wa/auth/refresh
POST	/api/wa/auth/logout
GET	/api/wa/config/extension
GET	/api/wa/health
POST	/api/wa/ingest
GET	/api/wa/conversations
GET	/api/wa/messages
GET	/api/wa/suggestions
POST	/api/wa/suggestions/:id/chosen
GET	/api/wa/quotes
POST	/api/wa/quotes/run
POST	/api/wa/conversations/:chat_id/upsert-lead
GET	/api/wa/followups
POST	/api/wa/followups
PATCH	/api/wa/followups/:id
POST	/api/wa/conversations/:chat_id/consent
POST	/api/wa/outbound
POST	/api/wa/outbound/:msg_id/confirm
POST	/api/wa/suggestions/run
POST	/api/wa/heartbeat

Método	Ruta
GET	/api/wa/operators
GET	/api/wa/metrics

webhooks.js — montado en (raíz)

Método	Ruta
POST	/ml
GET	/whatsapp
POST	/whatsapp

wolfboard.js — montado en /api/wolfboard

Método	Ruta
GET	/api/wolfboard/pendientes
POST	/api/wolfboard/sync
POST	/api/wolfboard/row
POST	/api/wolfboard/row-create
POST	/api/wolfboard/enviados
GET	/api/wolfboard/export
POST	/api/wolfboard/quote-batch

2. Rutas del frontend / SPA (27)

Ruta	Componente / página
*	→ redirect
/	PanelinCalculadora
/calculadora	PanelinCalculadora
/conductor	DriverTransportistaApp
/especificaciones	SpecManagementSandbox
/fichas	FichasPreview
/hub	BmcWolfboardHub
/hub/admin	AdminRoute
/hub/admin/analytics	AnalyticsModule
/hub/admin/users	UserAdminModule
/hub/agent-admin	AgentAdminModule
/hub/bugs	BugReportsList
/hub/canales	BmcCanalesUnificadosModule

Ruta	Componente / página
/hub/clientes	CientesMVP
/hub/cotizaciones	CotizacionesRoute
/hub/marketing	MarketingHubModule
/hub/ml	BmcMlOperativoModule
/hub/plan-import	BmcPlanImportModule
/hub/tareas	TasksModule
/hub/traktime/*	TraKtiMeModule
/hub/wa	BmcWaModuleWithTabs
/inspector	CalcLogicInspector
/logistica	BmcLogisticaApp
/mi-espacio	MySpacePage
/presentacion-licitacion	BidPresentation
/preview/pdf	PdfPreview
/wa	→ redirect

3. Integraciones y variables de entorno (67)

Extraído de `.env.example` (sólo nombres; nunca valores).

Prefijo	Variables
BMC_*	BMC_CALENDARIO_SHEET_ID · BMC_DASHBOARD_2_REPO · BMC_DEVELOPMENT_TEAM_REPO · BMC_EMAIL_INBOX_REPO · BMC_PAGOS_SHEET_ID · BMC_SHEET_ID · BMC_STOCK_SHEET_ID · BMC_VENTAS_SHEET_ID
ALERT_*	ALERT_CRITICAL_OFFLINE_RUNS · ALERT_CRITICAL_PCT · ALERT_EMAIL_FROM · ALERT_EMAIL_TO · ALERT_WARN_PCT
BUDGET_*	BUDGET_ENABLED · BUDGET_TOKENS_PER_24H · BUDGET_TURNS_PER_24H · BUDGET_TURNS_PER_5MIN · BUDGET_TURNS_PER_MIN
SHOPIFY_*	SHOPIFY_CLIENT_ID · SHOPIFY_CLIENT_SECRET · SHOPIFY_QUESTIONS_SHEET_TAB · SHOPIFY_SCOPES · SHOPIFY_WEBHOOK_SECRET
SMTP_*	SMTP_HOST · SMTP_PASS · SMTP_PORT · SMTP_SECURE · SMTP_USER
WOLFB_*	WOLFB_ADMIN_SHEET_ID · WOLFB_ADMIN_TAB · WOLFB_CRM_ENVIADOS_TAB · WOLFB_CRM_MAIN_TAB · WOLFB_DRY_RUN
GOOGLE_*	GOOGLE_APPLICATION_CREDENTIALS · GOOGLE_OAUTH_CLIENT_ID · GOOGLE_TASKS_CLIENT_ID · GOOGLE_TASKS_CLIENT_SECRET
WHATSAPP_*	WHATSAPP_ACCESS_TOKEN · WHATSAPP_APP_SECRET · WHATSAPP_PHONE_NUMBER_ID · WHATSAPP_VERIFY_TOKEN
IDENTITY_*	IDENTITY_COOKIE_DOMAIN · IDENTITY_COOKIE_NAME · IDENTITY_JWT_SECRET

Prefijo	Variables
KB_*	KB_ANALYTICS_LOG_MISS_QUESTION · KB_ANALYTICS_WINDOW_MAX_DAYS
ML_*	ML_CLIENT_ID · ML_CLIENT_SECRET
OPENAI_*	OPENAI_API_KEY · OPENAI_CHAT_MODEL
SHEETS_*	SHEETS_CLIENT_QUOTES_ENABLED · SHEETS_CLIENT_QUOTES_TAB
WA_*	WA_AUTH_EMAIL_FROM · WA_JWT_SECRET
AI_*	AI_GATEWAY_API_KEY
ANTHROPIC_*	ANTHROPIC_API_KEY
EMAIL_*	EMAIL_INGEST_TOKEN
FRONTEND_*	FRONTEND_BASE_URL
GEMINI_*	GEMINI_API_KEY
GROK_*	GROK_API_KEY
INTERNAL_*	INTERNAL_SUPERADMIN_EMAILS
MFA_*	MFA_KEY_HEX
PUBLIC_*	PUBLIC_BASE_URL
SUGGEST_*	SUGGEST_RESPONSE_USE_AGENT_CORE
SUPABASE_*	SUPABASE_PGP_ENCRYPT_KEY
SYNC_*	SYNC_HMAC_SECRET
VITE_*	VITE_GOOGLE_CLIENT_ID

4. Persistencia / migraciones (85 tablas)

migrations/ — 1 migración(es), tablas: `bmc_schema_migrations`, `quote_embeddings`

server/migrations/market-intel/ — 8 migración(es), tablas: `bmc_market_intel.competitors`, `bmc_market_intel.skus`, `bmc_market_intel.price_history`, `bmc_market_intel.etl_runs`, `bmc_market_intel.alerts`, `bmc_market_intel.mystery_shopping_queue`

shop-chat-agent/prisma/migrations/20240530213853_create_session_table/ — 1 migración(es)

shop-chat-agent/prisma/migrations/20250501044923_add_customer_tokens_table/ — 1 migración(es)

shop-chat-agent/prisma/migrations/20250502141909_add_code_verifier_table/ — 1 migración(es)

shop-chat-agent/prisma/migrations/20250508000001_add_conversation_tables/ — 1 migración(es)

shop-chat-agent/prisma/migrations/20251010121648_add_customer_account_urls_table/ — 1 migración(es)

supabase/migrations/ — 14 migración(es), tablas: `bmc_price_monitor.shopify_products`, `bmc_price_monitor.shopify_variants`, `bmc_price_monitor.ml_listings`, `bmc_price_monitor.search_keywords`, `bmc_price_monitor.ml_competitors`, `bmc_price_monitor.fx_settings`, `bmc_price_monitor.price_alerts`, `bmc_price_monitor.etl_runs`, `clientes.customers`, `clientes.customer_identities`, `clientes.customer_field_provenance`, `clientes.customer_events`, `clientes.customer_events_2026_04`, `clientes.customer_events_2026_05`, `clientes.customer_events_2026_06`, `clientes.customer_quotes`, `clientes.customer_purchases`, `clientes.customer_scores`, `clientes.customer_followups`, `clientes.automation_rules`, `clientes.customer_aliases`, `clientes.agent_jobs`, `clientes.agent_runs`, `archive.clientes_followups_2026_06`, `identity.message_threads`, `identity.message_thread_members`, `identity.messages`, `identity.user_activity_log`, `identity.users`, `identity.sessions`, `identity.modules`, `identity.role_grants`, `identity.module_grants`, `identity.access_requests`, `identity.quotes`, `identity.quote_events`, `identity.special_quote_requests`, `identity.notifications`, `identity.crm_personal_contacts`, `identity.crm_personal_leads`, `identity.audit_log`, `identity.mfa_secrets`, `tasks.task_lists`, `tasks.tasks`, `tasks.oauth_tokens`, `tasks.oauth_state`, `tasks.sync_log`, `tasks.sync_conflicts`, `public.oauth_states`

traktime-package/migrations/ — 11 migración(es), tablas: `tk_clients`, `tk_projects`, `tk_project_members`, `tk_tasks`, `tk_tags`, `tk_entries`, `tk_invoices`, `tk_invoice_lines`, `tk_invoice_seq`, `tk_audit_log`

transportista-cursor-package/migrations/ — 5 migración(es), tablas: `trips`, `trip_events`, `driver_sessions`, `outbox_notifications`

wa-package/migrations/ — 18 migración(es), tablas: `wa_conversations`, `wa_messages`, `wa_suggestions`, `wa_quotes`, `wa_followups`, `wa_heartbeats`, `wa_settings`, `wa_flags`, `wa_operators`, `wa_audit_log`, `wa_webhooks`, `wa_rules`, `wa_sla_breaches`, `bmc_quote_counter`

5. Auth & seguridad

- **Google OAuth** — `POST /api/auth/google` emite sesión (`server/routes/authGoogle.js`).

- **TOTP 2FA** — `POST /api/auth/mfa/challenge` (`server/routes/authMfa.js` + `server/lib/mfaTotp.js`).
- **Tokens JWT + refresh** — rotación obligatoria + reuse-detection (`server/lib/identityAuth.js`). Cookie `bmc_sess` (`httpOnly`, `SameSite=Strict`) sólo para `/api/auth/refresh` .
- **requireAuth** — valida el access-JWT (`server/middleware/requireAuth.js`).
- **RBAC por módulo** — grants `read / write / admin` (`server/middleware/requireGrant.js`).
- **Middleware:** `requireAuth.js` , `requireCrmCockpitAuth.js` , `requireGrant.js` , `requireServiceOrUser.js` , `requireWolfboardAuth.js` .

6. Scripts npm (209)

Script	Comando
<code>version:data</code>	<code>node scripts/update-calculator-data-version.js</code>
<code>quotation-preview:render</code>	<code>node scripts/render-quotation-preview-html.mjs</code>
<code>smoke:bmc-pdf</code>	<code>node scripts/bmc-pdf-export-smoke.mjs</code>
<code>preview:watch</code>	<code>node scripts/preview-dev.mjs</code>
<code>readme:generate</code>	<code>node scripts/generate-readme-presentation.mjs</code>
<code>readme:sync</code>	<code>npm run version:data && node scripts/generate-readme-presentation.mjs</code>
<code>readme:check</code>	<code>node scripts/generate-readme-presentation.mjs --check</code>
<code>disk:precheck</code>	<code>test "\$BMC_DISK_PRECHECK_SKIP" = "1" bash scripts/disk-space-precheck.sh .</code>
<code>predev</code>	<code>npm run disk:precheck</code>
<code>dev</code>	<code>npm run version:data && vite</code>
<code>dev:api</code>	<code>node --watch server/index.js</code>
<code>start:api</code>	<code>node server/index.js</code>
<code>dev:full</code>	<code>concurrently -n api,vite -c blue,green "npm run start:api" "npm run dev"</code>
<code>dev:full-stack</code>	<code>./run_full_stack.sh</code>
<code>local:view</code>	<code>bash scripts/local-view-autolaunch.sh</code>
<code>local:view:open</code>	<code>bash scripts/local-view-autolaunch.sh --open</code>
<code>local:stack:launchd:install</code>	<code>bash scripts/install-local-stack-launchagent.sh</code>
<code>local:stack:launchd:uninstall</code>	<code>bash scripts/uninstall-local-stack-launchagent.sh</code>
<code>dev:full:watch</code>	<code>concurrently -n api,vite -c blue,green "npm run dev:api" "npm run dev"</code>
<code>bmc-dashboard</code>	<code>node docs/bmc-dashboard-modernization/sheets-api-server.js</code>
<code>finanzas:inspect</code>	<code>node scripts/inspect-docker-context.cjs</code>

Script	Comando
finanzas:repro	bash scripts/repro-finanzas-404.sh
ml:cloud-run	./run_ml_cloud_run_setup.sh
ml:verify	bash scripts/verify-ml-oauth.sh
ml:corpus-export	node scripts/ml-export-full-corpus.mjs
wolfboard:run	bash scripts/wolfboard-run.sh
wolfboard:cotizar	node scripts/wolfboard-cotizar-batch.mjs
wolfboard:cotizar:force	node scripts/wolfboard-cotizar-batch.mjs --force
wolfboard:batch-ia	curl -s -X POST http://localhost:3001/api/wolfboard/quote-batch -H 'Content-Type: applicat
wolfboard:replay-fetch	node scripts/fetch-wolfboard-replay.mjs
ml:sim-batch	node scripts/ml-simulation-batch-export.mjs
ml:ai-audit	node scripts/ml-ai-audit-report.mjs
ml:pending-workup	node scripts/ml-pending-workup.mjs
etl:price-monitor	node scripts/price-monitor-etl.mjs
etl:price-monitor:dry	node scripts/price-monitor-etl.mjs --dry
proxy:openai	./run_proxy_openai.sh
prebuild	npm run disk:precheck && npm run version:data
build	vite build
preview	vite preview
tauri	npx --yes @tauri-apps/cli@^2
tauri:dev	npx --yes @tauri-apps/cli@^2 dev
tauri:build	npx --yes @tauri-apps/cli@^2 build
tauri:icon	npx --yes @tauri-apps/cli@^2 icon
demo:combinada-fijacion	node scripts/demo-combinada-fijacion-planta.mjs
docs:reference	node scripts/generate-system-reference.mjs
docs:tour	playwright test scripts/product-tour.spec.ts
docs:pdf	node scripts/build-product-pdf.mjs
docs:product	npm run docs:reference && npm run docs:tour && npm run docs:pdf
test	npm run test:core && npm run test:wa && npm run test:agent && npm run test:kb
test:core	node tests/validation.js && node tests/roofVisualQuoteConsistency.js && node tests/emailIn
test:wa	node tests/wa-ingest-contract.js && node tests/wa-enricher.test.js && node tests/wa-quote-
test:agent	node tests/agentTools.test.js && node tests/traktime-agent-tools.test.js && node tests/tra

Script	Comando
test:kb	node tests/kbSurfaceResolve.test.js && node tests/kbAnalytics.test.js && node tests/sugges
test:playwright:roof-bordes-local	node scripts/playwright-roof-bordes-local.mjs
test:playwright:hub-canales	node scripts/playwright-hub-canales-smoke.mjs
test:playwright:wizard	node scripts/playwright-wizard-happy-path.mjs
test:ml-auto	bash scripts/test-ml-auto-mode.sh
test:ml-auto:prod	bash scripts/test-ml-auto-mode.sh https://panelin-calc-q74zutv7dq-uc.a.run.app
test:e2e:wizard	node scripts/playwright-calculator-wizard.mjs
walkthrough:admin-cot	node scripts/playwright-admin-cot-walkthrough.mjs
walkthrough:admin-cot:headed	HEADED=1 node scripts/playwright-admin-cot-walkthrough.mjs
test:browser:full	node scripts/browser-test-suite.mjs
test:browser:full:local	node scripts/browser-test-suite.mjs --local
smoke:browser	node scripts/playwright-route-audit-smoke.mjs
smoke:browser:prod	node scripts/playwright-route-audit-smoke.mjs --base=https://calculadora-bmc.vercel.app
test:api	node tests/apiClient.test.js && node tests/operatorApiClient.test.js && node tests/teamAss
test:wa-integration	node tests/wa-enricher-integration.test.js
test:wa-pro	node tests/wa-config.test.js && node tests/wa-operator-auth.test.js && node tests/wa-rules
test:chat	node tests/chat-hardening.js
test:contracts	node scripts/validate-api-contracts.js
test:agent-golden	node tests/agentGolden/runner.mjs
mcp:panelin	node scripts/mcp-panelin-http.mjs
verify-tabs	node scripts/verify-sheets-tabs.js
setup-sheets-tabs	node scripts/setup-sheets-tabs.js
integrate-admin-cotizaciones	node scripts/integrate-admin-cotizaciones.js
watch-full-team	./scripts/watch-full-team-run.sh
map-all-sheets	node scripts/map-all-sheets-audit.js
launchpad	node scripts/launchpad-bmc-dev.js
launchpad:learn	node scripts/launchpad-bmc-dev.js --learn
go-live	./scripts/go-live-automation.sh
pre-deploy	./scripts/pre-deploy-check.sh

Script	Comando
verify:dev	bash scripts/bmc-dev-verify.sh
verify:dev:all	bash scripts/bmc-dev-verify.sh --all
verify:rdx	bash scripts/bmc-dev-verify.sh --all
gate:local	npm run lint && npm test && npm run test:api
dev:check:all	bash scripts/bmc-dev-check-all.sh
gate:local:full	npm run lint && npm test && npm run test:api && npm run build
check:env-drift	node scripts/check-env-drift.mjs
gate:secrets	node scripts/gate-secrets-drift.mjs
check	npm run gate:local
lint	eslint src/
team:hub	npx --yes serve docs -l 4710
open:email-env	bash scripts/open-email-inbox-env.sh
workspace:start	npm install && npm run env:ensure && node -e "console.log('\n=== Calculadora BMC workspace
workspace:autostart	bash scripts/workspace-folder-open.sh
env:ensure	bash scripts/ensure-env.sh
keys:audit	bash scripts/openai-key-audit.sh
keys:rotate	bash scripts/openai-key-rotate.sh
verify:google-drive-oauth	node scripts/verify-google-drive-oauth-env.mjs
verify:google-drive-dist	node scripts/verify-vite-google-client-in-dist.mjs
drive:bootstrap	bash scripts/drive-oauth-bootstrap.sh
drive:configure	node scripts/set-vite-google-client.mjs
drive:vercel-env	bash scripts/vercel-drive-env-push.sh
drive:one-shot	bash scripts/drive-oauth-one-shot.sh
panelsim:env	bash scripts/ensure-panelsim-sheets-env.sh
panelsim:email-ready	bash scripts/panelsim-email-ready.sh
panelsim:session	bash scripts/panelsim-full-session.sh
panelin:train:import	node scripts/panelin-training-import.mjs
capabilities:snapshot	node scripts/capabilities-snapshot.mjs
smoke:prod	node scripts/smoke-prod-api.mjs
smoke:routes	node scripts/demo-route-smoke.mjs
smoke:prod:no-ia	node scripts/smoke-prod-api.mjs --skip-suggest
bug-reports:setup-sheet	node scripts/setup-bug-reports-tab.mjs
bug-reports:ship	node scripts/bug-reports-ship-goal.mjs

Script	Comando
bug-reports:ship:bg	node scripts/bug-reports-ship-goal.mjs >> .runtime/goals/bug-reports-ship.log 2>&1 &
wa:cloud-check	node scripts/whatsapp-cloud-check.mjs
matriz:rename-headers	node scripts/matriz-rename-bromyros-headers.mjs
followup	node scripts/followup.mjs
program:status	node scripts/program-status.mjs
project:compass	node scripts/project-compass.mjs
schedule	npm run project:compass
channels:onboarding	node scripts/channels-onboarding.mjs
channels:automated	node scripts/channels-automated-pipeline.mjs
email:ingest-snapshot	node scripts/email-snapshot-ingest.mjs
matriz:reconcile	node scripts/reconcile-matriz-csv.mjs
matriz:reconcile-calc	node scripts/reconcile-calc-vs-matriz.mjs
matriz:bake	node scripts/bake-matriz-to-constants.mjs
matriz:pull-csv	node scripts/pull-matriz-csv.mjs
matriz:sync-fijaciones-isodec	node scripts/sync-fijaciones-isodec-bromyros.mjs
matriz:sync-silicona-300	node scripts/sync-silicona-300-neutra-bromyros.mjs
catalog:diff	node scripts/catalog-diff.mjs
catalog:diff:json	node scripts/catalog-diff.mjs --json
productos-maestro:reconcile	node scripts/reconcile-productos-maestro.mjs
productos-maestro:reconcile:json	node scripts/reconcile-productos-maestro.mjs --json
team:agent-matrix-skills	node scripts/merge-agent-matrix-skills.mjs
session:video-deps	bash scripts/user-session-video-deps.sh check
session:video-deps:ensure	bash scripts/user-session-video-deps.sh ensure
session:video-extract	bash scripts/user-session-video-extract.sh
session:video-ingest	bash scripts/user-session-video-ingest-from-iphone.sh
mac:storage-audit	bash scripts/mac-storage-audit-readonly.sh
mcp:cache:audit	bash scripts/mcp-cache-manager.sh audit
mcp:cache:prune	bash scripts/mcp-cache-manager.sh prune
mcp:cache:prune:apply	bash scripts/mcp-cache-manager.sh prune --apply
session:archive:audit	bash scripts/session-artifact-lifecycle.sh audit
session:archive:run	bash scripts/session-artifact-lifecycle.sh run
session:archive:run:prune	bash scripts/session-artifact-lifecycle.sh run --apply-prune

Script	Comando
session:archive:schedule:install	bash scripts/install-session-artifact-schedule.sh
session:archive:schedule:uninstall	bash scripts/uninstall-session-artifact-schedule.sh
session:archive:schedule:tick	bash scripts/session-artifact-scheduler.sh
transportista:migrate	node scripts/run-transportista-migrations.mjs
traktime:migrate	node scripts/run-traktime-migrations.mjs
wa:migrate	node scripts/run-wa-migrations.mjs
wa:admin	node scripts/wa-admin.mjs
wa:reconcile	node scripts/wa-reconcile-sheet.mjs
wa:purge-old	node scripts/wa-purge-old.mjs
wa:gen-docs	node scripts/wa-gen-config-docs.mjs
wa:ext:load	bash scripts/wa-extension-load.sh
wa:ext:rebuild	bash scripts/wa-extension-load.sh --rebuild
wa:ext:watch	bash scripts/wa-extension-load.sh --watch
visor:shopify-map	node scripts/build-quote-visor-shopify-map.mjs
visor:shopify-families	node scripts/build-quote-visor-shopify-families.mjs
visor:shopify-sync	npm run visor:shopify-map && npm run visor:shopify-families
panel:rendering:sync	node scripts/download-panel-rendering-assets.mjs
knowledge:scan	node scripts/knowledge-antenna-run.mjs --max-per-source=8 --min-score=0.45
knowledge:rank	node scripts/knowledge-antenna-rank.mjs
knowledge:impact	node scripts/knowledge-antenna-impact.mjs --days=14
knowledge:db	node scripts/knowledge-antenna-db.mjs --days=14
knowledge:direction	node scripts/knowledge-direction-tracker.mjs --max=12
training:report	node scripts/training-report.mjs
knowledge:report	node scripts/knowledge-antenna-run.mjs --max-per-source=8 --min-score=0.45
knowledge:env:ensure	bash scripts/knowledge-antenna-env-ensure.sh ensure
knowledge:env:check	bash scripts/knowledge-antenna-env-ensure.sh check
knowledge:env:cwd:guard	bash scripts/knowledge-antenna-env-cwd-regression.sh
knowledge:preflight	node scripts/knowledge-antenna-preflight.mjs --strict --sample=3
knowledge:magazine	node scripts/knowledge-antenna-magazine.mjs
knowledge:run	npm run knowledge:env:ensure && npm run knowledge:preflight && npm run knowledge:scan && n

Script	Comando
kb:build	node scripts/build-accessible-base.mjs
kb:build:quiet	node scripts/build-accessible-base.mjs --quiet
sheets:sync	node scripts/accessible-base-sync.js
sheets:sync:dry	node scripts/accessible-base-sync.js --dry-run
sheets:sync:watch	node scripts/accessible-base-sync.js --watch 3
sheets:sync:sheet	node scripts/accessible-base-sync.js --sheet
development:chain	node scripts/development-chain-runner.mjs
development:chain:full	node scripts/development-chain-runner.mjs --full
expert:workflow	node scripts/expert-dev-traceability.mjs workflow
expert:checkpoint	node scripts/expert-dev-traceability.mjs snapshot
expert:checkpoints	node scripts/expert-dev-traceability.mjs list
expert:restore-hint	node scripts/expert-dev-traceability.mjs restore-hint
knowledge:schedule:install	bash scripts/install-knowledge-antenna-schedule.sh
knowledge:schedule:uninstall	bash scripts/uninstall-knowledge-antenna-schedule.sh
knowledge:schedule:tick	bash scripts/knowledge-antenna-scheduler.sh
magazine:daily	node scripts/magazine-daily-digest.mjs
magazine:daily:send	node scripts/magazine-daily-digest.mjs --send
magazine:daily:dry	node scripts/magazine-daily-digest.mjs --dry-run
magazine:schedule:install	bash scripts/install-magazine-daily-schedule.sh
magazine:schedule:uninstall	bash scripts/uninstall-magazine-daily-schedule.sh
magazine:schedule:tick	bash scripts/magazine-daily-scheduler.sh
credentials:registry	node scripts/credentials-master-registry.mjs
credentials:registry:probe	node scripts/credentials-master-registry.mjs --probe-local
credentials:registry:encrypt	node scripts/credentials-master-registry.mjs --encrypt
credentials:registry:encrypt:only	node scripts/credentials-master-registry.mjs --encrypt-only
credentials:registry:decrypt	node scripts/credentials-master-registry.mjs --decrypt
test:playwright:admin-cot-suggest-ia	node scripts/playwright-admin-cot-suggest-ia.mjs
test:playwright:admin-cot-suggested-badge	node scripts/playwright-admin-cot-suggested-badge.mjs
test:playwright:admin-cot-wa-timeline	node scripts/playwright-admin-cot-wa-timeline.mjs

Script	Comando
test:playwright:admin-cot-fl-completion	node scripts/playwright-admin-cot-fl-completion.mjs
test:market-intel	node --test tests/market-intel/*.test.js
migrate:market-intel	node server/migrations/market-intel/run-migrations.js
etl:run	node -e "require('./server/lib/marketIntel/scheduler')"

7. CI / workflows (11)

Workflow	Nombre	Disparadores
catalog-diff.yml	Catalog ↔ MATRIZ diff	schedule, pull_request, workflow_dispatch
ci.yml	CI — Panelin Calculadora BMC	push, pull_request, workflow_dispatch, schedule
deploy-calc-api.yml	Deploy Calculator API to Cloud Run	workflow_dispatch
deploy-vercel.yml	Deploy Frontend to Vercel	pull_request, workflow_dispatch
drive-oauth-dist-verify.yml	Drive OAuth — verify Client ID in dist	workflow_dispatch
drive-oauth-verify.yml	Drive OAuth — verify Client ID	workflow_dispatch
email-ingest-scheduled.yml	Email ingest — scheduled	workflow_dispatch, schedule
knowledge-antenna-reusable.yml	Knowledge Antenna reusable	workflow_call
knowledge-antenna-scheduled.yml	Knowledge Antenna — scheduled	workflow_dispatch, schedule
product-docs.yml	Product Docs (auto-update)	schedule, workflow_dispatch
smoke-prod-scheduled.yml	Smoke — prod API (scheduled)	schedule, workflow_dispatch

8. Agentes Claude (12)

- **bmc-api-contract** — "Validates API responses against canonical contracts for the BMC project. Detects drift between server routes and expected shapes before the UI breaks. Use when
- **bmc-calc-specialist** — "Specialist for the Panelin Calculator (calculadora-bmc.vercel.app, port 5173). Knows BOM, pricing logic, panel constants, techo/pared calculations, Drive integ
- **bmc-deployment** — "Deployment specialist for calculadora-bmc.vercel.app (Vercel frontend) and panelin-calc Cloud Run (Node.js API). Handles deploy, rollback, diagnostics, env var
- **bmc-docs-sync** — "Keeps [PROJECT-STATE.md](#), docs/, and propagation protocol up to date for the BMC project. Detects stale docs, broken links, missing README entries, and agents th
- **bmc-fiscal** — "Fiscal oversight and operational efficiency agent for BMC Uruguay (SAS). Detects IVA/IRAE/BPS inconsistencies, monitors PROJECT-STATE protocol compliance by ot

- **bmc-judge** — "Evaluates team work quality and agent performance for BMC/Panelin project. Generates run reports, historical rankings, and next-step recommendations. Use when
- **bmc-orchestrator** — "Orchestrates full BMC team runs for the calculadora-bmc.vercel.app project. Coordinates all roles in order, applies Run Scope Gate, manages handoffs, and itera
- **bmc-panelin-chat** — "Specialist for the Panelin AI chat system and Developer Training Mode. Knows PanelinChatPanel, PanelinDevPanel, useChat hook, agentChat SSE endpoint, training
- **bmc-panelin-mcp** — External agent surface for the Panelin BMC calculator. Exposes 22 tools (calc + catalog + state + PDF + CRM + WhatsApp + telemetry) over MCP for use by Claude C
- **bmc-security** — "Security reviewer for BMC/Panelin project. Audits OAuth tokens, env vars, CORS, HMAC webhooks, credentials, and pre-deploy security checklist. Use when asked f
- **bmc-sheets-mapping** — "Google Sheets and planilla mapping specialist for BMC. Knows CRM_Operativo, Master_Cotizaciones, Pagos_Pendientes, Metas_Ventas, AUDIT_LOG schemas, column mapp
- **calculo-especialist** — "Specialist for the 2D roof plan SVG dimensioning system (planta de cotas). Knows buildPanelLayout, PanelChainDimensions, PanelLabels, VerificationBadge, roofPI

Regenerar: `node scripts/generate-system-reference.mjs`.